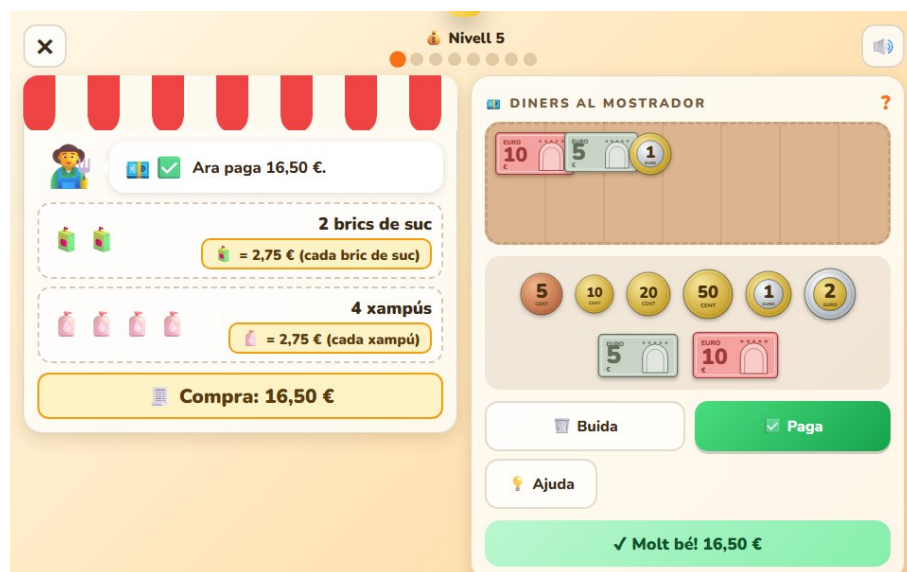


Al Mercat!

Maths with euro coins and notes · Age 12-13 and newcomer classes



What it teaches

Addition, subtraction and multiplication with euros, simple decimals (€1.25) and prices per kilo, in a real context: shopping at a market. Students pay by putting coins (1 cent to €2) and notes (€5 to €50) on the counter, and in the change levels they play the shopkeeper and give the coins back to the customer.

The 10 levels

Level	Content	Example
1. A euro each	What you do: Buy 2 to 5 identical products and pay with €1 and €2 coins. What you learn: Multiplying is adding many times: $5 \times €1 = €5$.	5 apples · €1 each = 5 €
2. Whole prices	What you do: Buy products up to €10 and pay with coins and notes. What you learn: Multiplying with bigger prices: $3 \times €2 = €6$.	3 milks · €2 each = 6 €
3. The shopping list	What you do: Type the total of a list of 2 or 3 products, then pay. What you learn: Adding prices and multiplying: $2 \times €1 + 1 \times €2 = €4$.	$2 \times 1 € + 1 \times 2 € = 4 €$
4. Half a euro	What you do: Prices with half euros (€0.50, €1.50...). Type the total and pay with 50-cent coins. What you learn: Easy decimals: $3 \times €1.50 = €4.50$.	3 biscuits · 1,50 € = 4,50 €
5. Cents	What you do: Prices like €1.25 or €0.75. Type the total and pay with cent coins. What you learn: Adding and multiplying with cents: $3 \times €0.75 = €2.25$.	3 apples · 0,75 € = 2,25 €
6. Change	What you do: You are the shopkeeper! The customer pays with a note: give the change in coins. What you learn: Subtracting: $€5 - €3.50 = €1.50$.	Total 3,50 € · pays 5 € → change 1,50 €
7. Buy and change	What you do: Work out the total of a purchase, then give the customer their change. What you learn: Adding, multiplying and subtracting, all together.	Total 6,75 € · pays 10 € → change 3,25 €

Level	Content	Example
8. Price per kilo	What you do: Buy products by weight (1, 2 or 3 kg). Type the total and pay. What you learn: Price per kilo: $2 \text{ kg} \times €3 = €6$.	$2 \text{ kg apples} \cdot 3 \text{ €/kg} = 6 \text{ €}$
9. Half a kilo	What you do: Weigh half a kilo or a kilo and a half. Type the total and pay. What you learn: Half a kilo is half the price.	$0,5 \text{ kg} \cdot 3 \text{ €/kg} = 1,50 \text{ €}$
10. Quarter kilos	What you do: Weigh 250 g or 750 g. Type the total and pay. What you learn: 250 g is a quarter of a kilo: the price $\div 4$.	$250 \text{ g} \cdot 4 \text{ €/kg} = 1 \text{ €}$

How to use it in class

- Open robertpotau.github.io/games/al-mercat/ on a computer, tablet or phone. Nothing to install and no sign-up.
- Each student creates a profile (name, avatar) and picks a language: **Catalan, Spanish or icons only + voice** (the task is read aloud).
- Levels can be chosen freely; the “Daily challenge” mixes purchases from different levels. The helper counter is automatic on levels 1-4 (change it in the settings cog).
- **Teacher panel:** 5 quick clicks on the “Al Mercat!” title. It shows each student's progress, lets you change the purchases per game and exports the data to CSV.
- Progress is stored in the device's browser; no data leaves the computer.

Tip: for students who have just arrived, start at level 1 in “Icons only + voice” mode and keep the helper counter on.